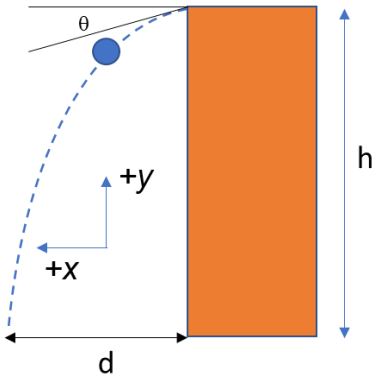
The next four questions pertain to the situation described below.

A ball with mass m =1.8 kg is thrown downward from the top of a building with height h = 16 m and initial speed v_0 at an angle θ with respect to the horizontal as shown in the figure. The x-component of the initial velocity v_{x0} = 23.4 m/s, and the ball hits the ground a distance d = 24 m from the building as shown in the figure.



1) How long is the ball in the air?

a. t = 1.81 s
b. t = 1.03 s
c. t = 4.78 s

$\Delta x = d = v_{ox} t$

$t = \frac{d}{v_{ox}} = \frac{24}{23.4} = 1.03s$

2) What is v_{y0}, the y component of the ball's initial velocity?

a. v_{y0} = -19 m/s
b. v_{y0} = -13 m/s
c. v_{y0} = -26 m/s
d. v_{y0} = -11 m/s
e. v_{y0} = 0 m/s

$\Delta y = v_{oy} t - \frac{1}{2} g t^2 \Rightarrow v_{oy} = \frac{\Delta y + \frac{1}{2} g t^2}{t}$

$v_{oy} = \frac{-16 + \frac{1}{2}(9.8)(1.03)^2}{1.03}$

$v_{oy} = -11 \text{ m/s}$

3) If the ball is thrown at the same angle, but with a greater speed, the ball would be in the air

a. a **shorter** time.
b. a **longer** time.
c. the **same** amount of time.

v_0 vs v_0 at angle $\quad v_{oy} = v_0 \sin\theta$

v_{oy} will be bigger!! Shorter time

4) If the ball is thrown with the same speed, but horizontally (e.g. θ=0), it would hit the ground with

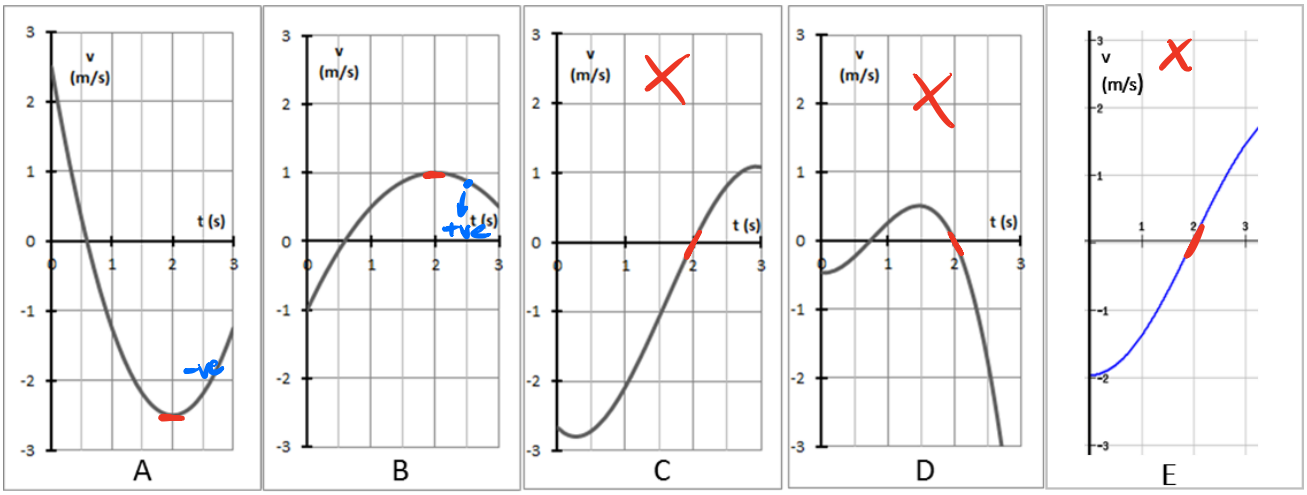
a. a **faster** speed than when it was thrown downward.
b. a **slower** speed than when it was thrown downward.
c. the **same** speed as when it was thrown downward.

$v_x = v_0 \qquad v_y = \sqrt{2gh}$

$v = \sqrt{v_0^2 + 2gh}$

velocity vs time graphs

The next two questions pertain to the situation described below.



A ball's motion is represented by the velocity time-graph above.

5) *Which of the velocity-time graphs best represents this statement?*

An object has zero acceleration at t=2 s. At t=2.5 s, it is moving in the positive direction and slowing down.

$a = \frac{dv}{dt} \Rightarrow$ slope = 0

V is +ve

a. A
b. B
c. C
d. D
e. E

eliminate C, D, E

6) *Which of the velocity-time graphs best represents this statement?*

The object is moving in the positive direction and speeding up at t=1.75 s. The object has has negative acceleration at t=2.5 s.

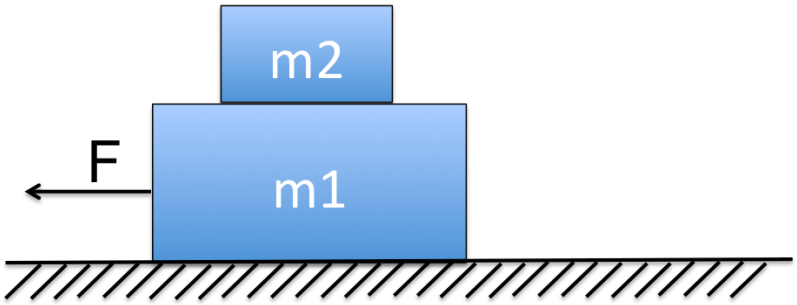
eliminate A ⇒ V is -ve

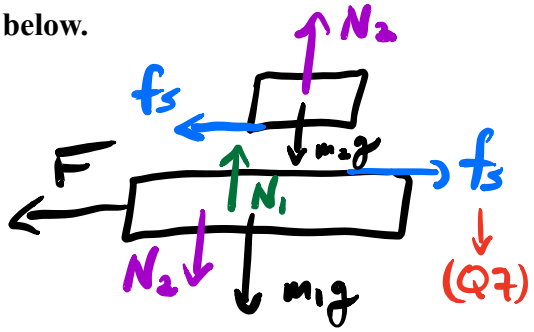
a. A
b. B
c. C
d. D
e. E

B is speeding up @ 1.75
positive direction

@ 2.5s it's slowing down a is -ve

The next three questions pertain to the situation described below.





A box of mass $m_1 = 10$ kg is placed on a frictionless surface. A smaller box of mass $m_2 = 3$ kg is placed on top of the larger box. The system of boxes is initially at rest. A horizontal force F is applied to the lower box, pulling it to the left as shown in the figure. The coefficients of static and kinetic friction between the two boxes are $\mu_k = 0.25$ and $\mu_s = 0.29$.

7) Initially, the two boxes are observed to move together (the small box remains on top of the large box). What is the direction of the friction force applied **by** the smaller box m_2 **on** the larger box m_1?

a. to the left
b. to the right
c. there is no frictional force between the two boxes

8) What is the maximum force F_{max} that can be applied, before the small box begins to slide relative to the large box?

a. $F_{max} = 37$ N
b. $F_{max} = 7.4$ N
c. $F_{max} = 32$ N
d. $F_{max} = 28$ N
e. $F_{max} = 24$ N

$f_{s,max} = \mu_s N_2 = \mu_s m_2 g = m_2 a \quad (1)$

$m_1: \; F - f_s = m_1 a \quad (2)$

eliminate a from (1) $a = \mu_s g$

$F = m_1 a + f_s = \mu_s m_1 g + \mu_s m_2 g = \mu_s (m_1 + m_2) g$

$F = 0.29(10+3)(9.8) \Rightarrow \boxed{F = 37\text{N}}$

9) If the boxes start from rest, and a force $F = 12$ N is applied until the boxes have moved a distance of 2.5 m, what is the work done by friction on the **top** box m_2? (You may assume the two boxes move together.)

a. $W_{friction} = 6.9$ J
b. $W_{friction} = 0$ J
c. $W_{friction} = 21$ J
d. $W_{friction} = -6.9$ J
e. $W_{friction} = -21$ J

$a = \frac{F}{m_1 + m_2} = \frac{12}{10+3} \Rightarrow a = 0.923\ m/s^2$

$f_s = m_2 a$

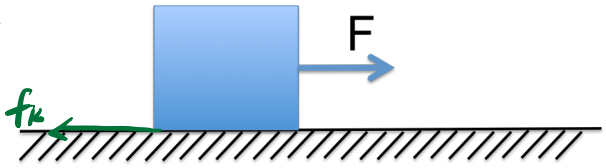
$f_s = 3(0.923) = 2.77\text{N}$

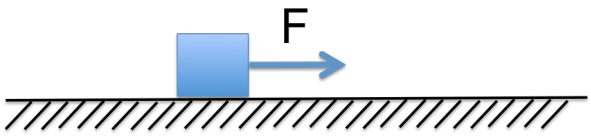
$W_{f_s} = f_s \cdot \Delta x \cdot \cos\theta = 2.77(2.5)\cos 0$

$\boxed{W_{f_s} = +6.9\text{J}}$

The next two questions pertain to the situation described below.

A solid cubic box of mass $m = 4$ kg is pushed along a floor by a horizontal force $F = 12$ N. The box is observed to move with a constant velocity $v = 0.3$ m/s.





10) What is the coefficient of kinetic friction between the box and the floor?

a. μ= 0.0092
b. μ= 0.15
c. μ= 0.31

$F - f_k = 0$

$f_k = F$

$\mu_k mg = F$

$\mu_k = \frac{F}{mg} = \frac{12}{4(9.8)} = 0.31$

11) A second box is made of the same material as the first box but is half the size of the first box in every dimension. Subject to the same force $F = 12$ N, the second box:

a. Moves with the same constant velocity as the first box.
b. Moves with a constant acceleration
c. Moves with a constant velocity eight times that of the first box.

Volume $= \frac{V_0}{8}$; ρ = same

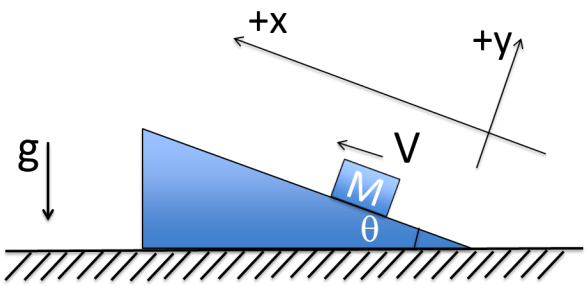
$m_0 = \rho V_0$

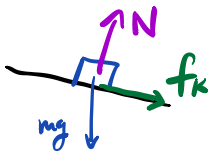
$m' = \rho \frac{V_0}{8} = \frac{m_0}{8} = \frac{4}{8} = 0.5\,kg$

$f_k = \mu_k mg$

f_k is smaller; box will accelerate

The next three questions pertain to the situation described below.





A box of mass $M = 0.4$ kg is moving along a stationary ramp upward ($+x$ direction) as shown in the figure. The box's initial speed $V = 1.4$ m/s. The angle between the ramp and the horizontal plane $\theta = 24$ degrees, and the coefficient of friction between the box and the ramp is $\mu_k = 0.32$.

12) What is the magnitude of the box's acceleration as it is moving up the ramp?

a. $|a_{up}| = 6.9$ m/s^2
b. $|a_{up}| = 7.7$ m/s^2
c. $|a_{up}| = 1.1$ m/s^2
d. $|a_{up}| = 0.85$ m/s^2
e. $|a_{up}| = 4$ m/s^2

$\Sigma F = ma = -mg\sin\theta - f_k = -mg\sin\theta - \mu_k mg\cos\theta$

$a = -g(\sin\theta + \mu_k \cos\theta)$

$a = -9.8(\sin 24 + 0.32\cos 24)$

$a = -6.9\ m/s^2$

$|a| = 6.9\ m/s^2$

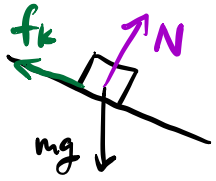
13) What is the direction of the box's acceleration as it moves up the ramp?

a. $+x$
b. $-x$
c. 0

acceleration is down the ramp.

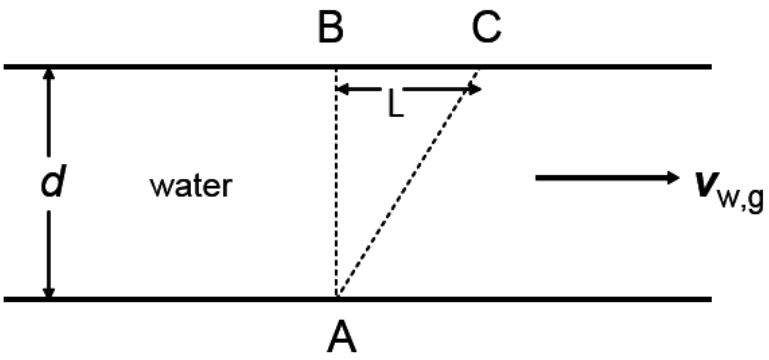
14) Eventually the box is observed to slide back down the ramp with constant acceleration a_{down}. Compare the magnitude of the acceleration of the box going up and down the ramp.

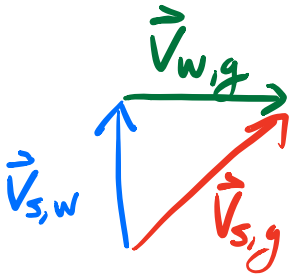
a. $|a_{down}| < |a_{up}|$
b. $|a_{down}| = |a_{up}|$
c. $|a_{down}| > |a_{up}|$



- friction is opposite of $W_{\parallel} = mg\sin\theta$
- acceleration will have a smaller magnitude

The next two questions pertain to the situation described below.





The width of a river is $d = 90$ m, the water flows with a constant speed $v_{w,g}$ with respect to its banks. An athletic swimmer can swim at a speed $v_{s,w} = 6$ m/s in still water. Although she heads directly across the river toward the opposite bank with her maximum effort, she arrives at point **C** on the opposite bank a distance $L = 60$ m downstream.

15) The speed of the water, relative to the ground is:

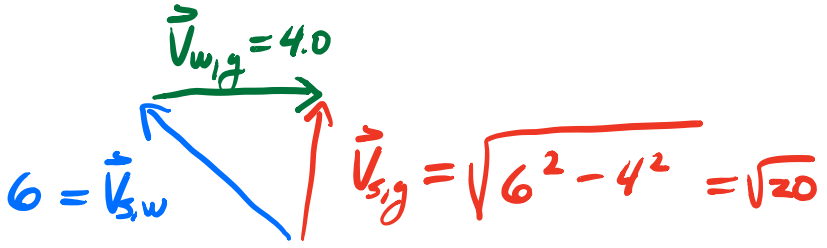
a. $v_{w,g} = 7.2$ m/s
b. $v_{w,g} = 3$ m/s
c. $v_{w,g} = 6$ m/s
d. $v_{w,g} = 4$ m/s
e. $v_{w,g} = 10$ m/s

$V_{w,g} = \frac{L}{t} \Rightarrow$ need to find time

$V_{s,w} = \frac{d}{t} \Rightarrow t = \frac{d}{V_{s,w}}$; $V_{w,g} = \frac{L}{d} V_{s,w} = \frac{60}{90} \cdot 6 = 4\,m/s$

16) If instead she chose a heading that would get her directly across the river (point **B**), how long would it take her to get to the other side?

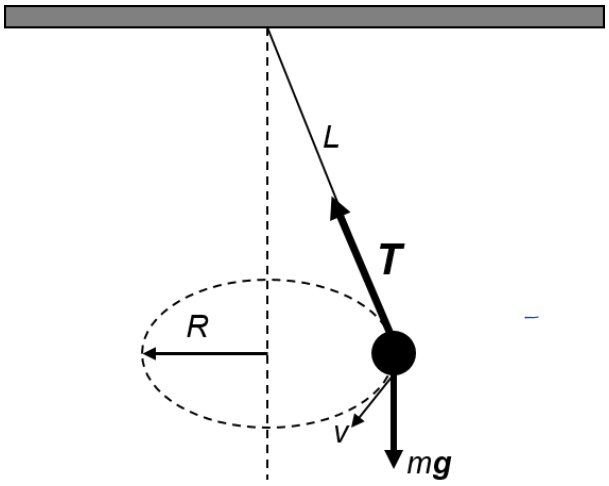
a. t = 45 s
b. t = 20 s
c. t = 15 s



$V_{s,g} = \frac{d}{t} \Rightarrow t = \frac{d}{V_{s,g}} = \frac{90}{\sqrt{20}} = 20s$

The next two questions pertain to the situation described below.

A ball of mass mass m = 1.5 kg is suspended from a rope of length L and travels in a horizontal circle of radius R = 1.8 m at a constant speed v = 3 m/s, as shown in the figure.



17) What is the horizontal component of the tension in the rope $T_{\text{horizontal}}$?

a. $T_{\text{horizontal}}$ = 15 N
b. $T_{\text{horizontal}}$ = 7.5 N
c. $T_{\text{horizontal}}$ = 8.3 N

$$T_h = m a_c = \frac{m v^2}{R}$$

$$T_h = \frac{1.5\,(3)^2}{1.8} = 7.5\text{N}$$

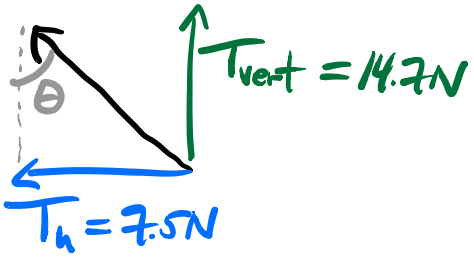
18) What is the length L of the rope?

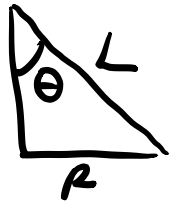
a. L = 2.02 m
b. L = 3.96 m
c. L = 2.42 m

$$T_{vert} = mg = 1.5(9.8) = 14.7\text{N}$$

$$\theta = \tan^{-1}\left(\frac{T_h}{T_{vert}}\right)$$

$$\theta = \tan^{-1}\left(\frac{7.5}{14.7}\right) = 27.0^\circ$$





$$\sin\theta = \frac{R}{L} \Rightarrow L = \frac{R}{\sin\theta} = \frac{1.8}{\sin 27} = 3.96\text{ m}$$

The next two questions pertain to the situation described below.

For the following problems assume the earth is a sphere with radius $R_E = 6.4 \times 10^6$ m, and mass $M_E = 6 \times 10^{24}$ kg.

19) If a person weighs 120 pounds on the surface of the Earth, how much would the person weigh on the surface of a planet that has 1/2 of the mass of the earth and 1/2 of the Earth's radius?

a. 120 pounds
b. 60 pounds
c. 240 pounds



$M_x = \frac{1}{2} M_{Earth}$ $R_x = \frac{1}{2} R_E$

$g = \frac{GM}{R^2}$ $g_x = 2 g_{earth}$

20) What is the speed of a 128 kg satelite in a circular orbit around the earth at a distance of $R_{orbit} = 1.92 \times 10^7$ m from the center of the earth?

a. 0.00808 m/s
b. 4570 m/s
c. 1250 m/s
d. 7910 m/s
e. 1.12×10^5 m/s

$$\frac{mv^2}{R} = \frac{GmM}{R^2}$$

$$V = \sqrt{\frac{GM}{R}} = \sqrt{\frac{6.67 \times 10^{-11} \cdot (6.0 \times 10^{24})}{1.92 \times 10^7}}$$

$$V = 4570 \text{ m/s}$$